

DevOps, Software Evolution and Software Maintenance, MSc

Course Code: KSDSESM1KU

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1. System's Perspective

1.1 Design and Architecture

When tasked with switching to another language for the system, C# was chosen. Accordingly, the project was built with Razor Pages and Entity Framework Core. The choice of C# was due to much of the group's pre-existing familiarity with the language, which streamlined the development process, especially when it came to the architecture.

The architecture of the project follows a layered onion architecture split into three parts: Core, Infrastructure, and Web. The below visualization shows the responsibilities of each layer:

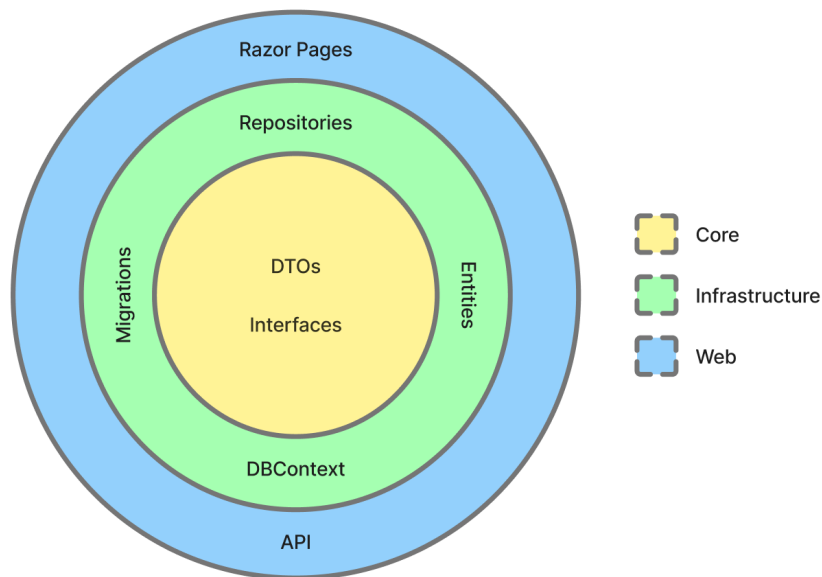


Figure 1: System Architecture

- The Core layer of the system focuses on handling DTOs and repository interfaces. This layer does not reference any frameworks or libraries, staying independent from the rest of the system.
- The Infrastructure layer focuses on the database context, migrations, and the implementation of repository interfaces. This layer depends only on the Core, whilst having no reference to the Web layer.
- The Web layer handles the UI through Razor Pages along with the API. It also acts as the base of the system, handling the dependency injection and referencing both the Core and Infrastructure layers of the application.

1.1.1 Choice of Final Infrastructure-as-Code Architecture We ended up migrating to Terraform towards the end of the project as it allowed for easy maintenance and resource control through defined interfaces. Terraform has a thoroughly defined Documentation for Digital Ocean resources, and the migration to defining existing Vagrant deployments along with “Click-Ops” resources therefore did not have much extra overhead.

1.2 Dependencies of MiniTwit

- **Git / GitHub** (*Development, CI/CD*) Source control, reviews, and workflow hosting
- **Trello** (*Development, operations*) Backlog management and work tracking
- **Discord** (*Development, operations*) Team communication and receiving alerts (for example from GitHub & Grafana webhooks)
- **C# / .NET** (*Development, production*) Application language and runtime
- **NuGet** (*Development, CI/CD*) Package restore and feeds for .NET dependencies
- **ASP.NET Core (Razor Pages)** (*Development, production*) Web UI and HTTP API
- **Entity Framework Core** (*Development, production*) Database access and migrations
- **PostgreSQL** (*Development, testing, production*) Primary data store (managed in DigitalOcean)
- **Docker** (*Development, CI/CD, production*) Container images and runtime isolation
- **Docker Hub** (*CI/CD, production*) Registry for built application images
- **Docker Compose** (*Development, testing*) Local and test multi-container setups
- **Docker Swarm** (*Production*) Orchestration and rolling updates
- **DigitalOcean** (*Infrastructure, production*) Cloud VMs, managed database, and networking
- **DigitalOcean Spaces (S3-compatible)** (*Infrastructure, CI/CD*) Object storage with an S3-compatible API (Terraform remote state backend)
- **Terraform** (*Infrastructure, CI/CD*) Infrastructure as code for managing cloud resources
- **GitHub Actions** (*CI/CD*) Continuous integration and deployment pipelines
- **Third-party GitHub Actions** (*CI/CD*) Marketplace and vendor-maintained workflow steps (for example checkout, Docker login, Terraform setup, PR plan commenter, GitHub App token)
- **GitHub CLI** (*CI/CD*) Command-line GitHub integration in workflows (for example `gh auth setup-git` for automated commits)
- **Ubuntu** (*CI/CD, production*) Base operating system on GitHub-hosted runners and provisioned droplets
- **SSH (OpenSSH)** (*CI/CD, production*) Remote deploy, file copy, and server access from pipelines

- **Prometheus** (*Monitoring*) Metrics collection
- **Grafana** (*Monitoring*) Dashboards and alerting
- **Loki** (*Monitoring*) Centralized log storage
- **Promtail** (*Monitoring*) Shipping container logs to Loki
- **Python** (*Testing (local and CI/CD)*) API simulator test driver and test scripts
- **Selenium** (*Testing (local and CI/CD)*) Browser-based UI tests (with Chrome in Docker)
- **dotnet format** (*Development, CI/CD*) C# formatting and verifying the tree matches the formatter in CI
- **Roslynator** (*Development, CI/CD*) C# static analysis (Roslyn-based diagnostics)
- **Codespell** (*Development, CI/CD*) Spell checking across the repository
- **Hadolint** (*Development, CI/CD*) Linting Dockerfiles
- **Codacy** (*Development, CI/CD*) Hosted static analysis and pull-request quality checks
- **SonarCloud** (*Development, CI/CD*) SonarQube-family analysis and quality gate on pull requests
- **CodeQL** (*Development, CI/CD*) Semantic security and quality scanning (for example C# and Python) on pull requests
- **Docker Scout** (*CI/CD*) Container image vulnerability scanning on QA builds
- **OpenAPI Generator** (*Development*) Generating the API simulator stub from the OpenAPI description
- **Pandoc** (*Report CI/CD*) Converting the report Markdown to PDF in automation
- **LaTeX (pdflatex / TeX Live)** (*Report CI/CD*) PDF engine used by Pandoc for report builds
- **librsvg (rsvg-convert)** (*Report CI/CD*) Converting committed SVG figures (for example Mermaid diagrams) for inclusion in the PDF
- **GNU Make** (*Development, CI/CD*) Task automation (local and in workflows)

Libraries NuGet references come from the Razor Pages solution (`razor-pages/Web`, `razor-pages/Infrastructure`). Direct Python libraries for automated tests are listed below; the UI test lockfile also pins transitive versions (see `tests/selenium/requirements.txt`).

- **Microsoft.AspNetCore.Identity.EntityFrameworkCore** (*Web, Infrastructure*) ASP.NET Core Identity integrated with EF Core
- **Microsoft.EntityFrameworkCore.Design** (*Web, Infrastructure*) EF Core design-time support and migrations
- **Npgsql.EntityFrameworkCore.PostgreSQL** (*Web, Infrastructure*) EF Core provider for PostgreSQL
- **Newtonsoft.Json** (*Web*) JSON serialization and deserialization
- **Swashbuckle.AspNetCore.Annotations** (*Web*) OpenAPI metadata

and attributes for the HTTP API

- **Swashbuckle.AspNetCore.Newtonsoft** (*Web*) OpenAPI generation using Newtonsoft.Json
- **DotNetEnv** (*Web*) Loading environment variables from `.env` files
- **prometheus-net.AspNetCore** (*Web*) Prometheus metrics for ASP.NET Core
- **Microsoft.CodeAnalysis.Analyzers** (*Infrastructure*) Build-time Roslyn analyzers
- **Microsoft.Extensions.Hosting** (*Infrastructure*) Hosting abstractions for background-style infrastructure code
- **prometheus-net** (*Infrastructure*) Prometheus metric registration and exposition primitives
- **Npgsql** (*Infrastructure*) PostgreSQL data provider (ADO.NET) alongside EF
- **TimeZoneConverter** (*Infrastructure*) Resolving time zones in infrastructure logic
- **requests** (*tests/API_Spec*) HTTP calls from the API simulator
- **pytest** (*tests/selenium*) Test runner for the Selenium UI suite
- **selenium** (*tests/selenium*) WebDriver client driving the remote Chrome grid

1.3 Current State of Our Systems

The current state of our system leaves it steadily functional across performance, scalability, code quality, security, and testing. However, there are still some limiting factors that should be dealt with prior to considering it a complete product.

In regards to performance and scalability, the primary issue we encountered is the resources available to the server being insufficient for exceptionally high traffic. However, this was mainly attributed to the DigitalOcean server plan tier we used rather than a structural limitation of the application itself.

The test coverage is fairly extensive across the API and browser-based UI levels, but there could be more explicit tests for base application logic, as well as error and edge-case interactions and security behavior.

Static Analysis and Code Quality Tools

- **SonarQube** indicates a few potential reliability and maintainability issues, as well as some security hotspots, but still gives it an A-rating in the main issue categories.
- **Codacy** gives the application as a whole an A-rating, but also indicates a few potential security hotspots.
- **CodeQL** passes on all its vulnerability checks.
- **Hadolint** and **Roslynator** show no issues.

SonarQube's Quality Assessment:

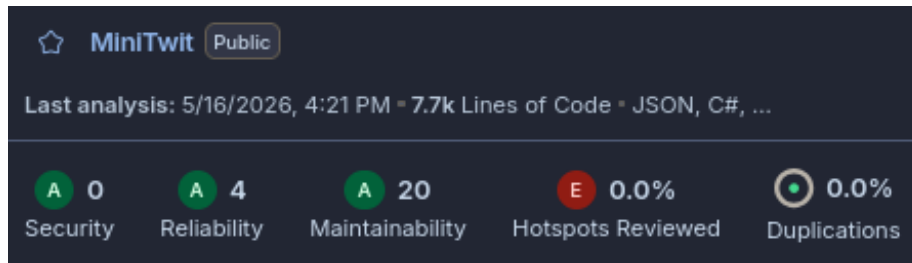


Figure 2: SonarQube Quality Assessment

The issues mainly consist of maintainability and style problems, such as inconsistent naming, improper exception handling, and potential accessibility and configuration problems. Additionally, there are a few issues involving some incomplete implementations and asynchronicity handling.

2. Process' perspective

2.1 CI/CD pipelines, deployment, and release

All development work is done on branches and requires a pull request to be merged into the main branch. Pull requests are automatically checked with code scanning tools and also triggers a QA build which runs a full build, test and deployment to a QA droplet and database. Note that due to limitations on number of allowed droplets in our Digital Ocean account level, this QA droplet was later included in the production swarm as well.

After merging a pull request into main, the report pdf is built if changes have been made in the relevant files. Nothing is immediately pushed to production as we deemed that we wanted our releases to contain more than a single small change, as well as to have more control of when releases to production were made. The control of timing is important to ensure stability of the application and timely action given a failure/bug.

We used an automated pipeline to deploy our production services that automatically triggers when a tag is pushed to the repository. We attempted to follow a form of semantic versioning (<https://semver.org/>) for tag names in order to have a consistent format and a notion of how big each release was. The automated deployment builds a docker image and deploys the stack on the swarm leader node.

Monitoring is deployed manually in a separate workflow. The monitoring droplet was initially a stand-alone droplet, but given the Digital Ocean limitation on droplets, this droplet was also later included in the swarm. The monitoring deployment could have been automatically deployed if changes appeared in the relevant root folder, yet changes to the configurations were rather rare and we therefore did not deem it necessary.

Below is an overview of the different stages of development towards operationalization. In the following sections we will deep dive into the QA deployment workflow, continuous deployment release workflow, and the monitoring deployment workflow.

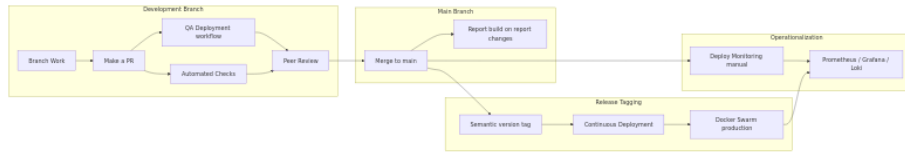


Figure 3: End-to-end flow chart for CI/CD

Pull-request pipeline (QA Deployment) The QA deployment is defined in [.github/workflows/continous-QA-deployment.yaml](#) and is automatically run on pull requests towards the main branch.

The above flowchart shows the various steps and interactions between systems happening during the QA Deployment and test workflow. The workflow runs at the same time as the static code analysis tools: **CodeQL**, **SonarCube**, and **Codacy**.

Production release (Continuous Deployment) The Continuous deployment to production is defined in [.github/workflows/continous-deployment.yaml](#) and is automatically run on tags pushed to the main branch.

Monitoring deployment The monitoring stack deployment is defined in [.github/workflows/monitor-deployment.yaml](#) and runs only when someone manually triggers it.

Deployment & Release Summary

- **Local** (*via `make app-build` — `compose-test.yaml`, port 8081*) Using Local Docker Compose
- **QA (pre-merge)** (*QA Deployment workflow on pull request*) With Docker Hub image `testminitwit:latest`, Using Compose on test droplet
- **Production** (*tag, then Continuous Deployment*) With Docker Hub image `minitwitimage:<sha>`, Using Docker Swarm (3 replicas)
- **Monitoring** (*manual Deploy Monitoring workflow*) Using Swarm stack monitoring

2.2 Monitoring

The monitoring of our application is done through the use of Prometheus and Grafana.

The data collection is handled by Prometheus' .NET client library, `prometheus-net`, where `UseMetricServer` collects and exposes metrics for use by Grafana.

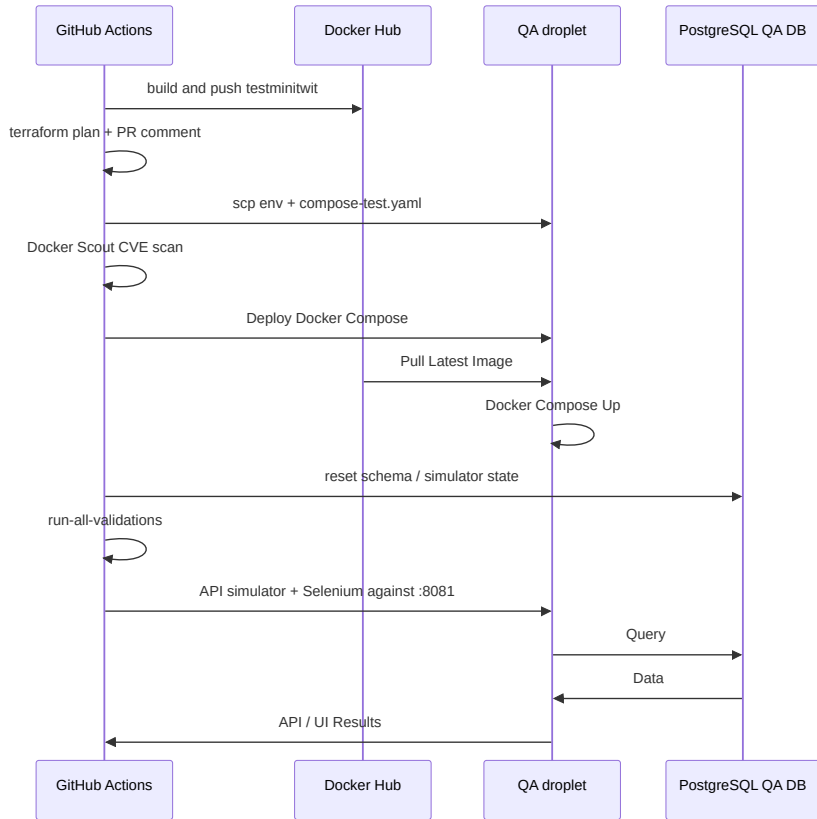


Figure 4: Complete QA Build Workflow

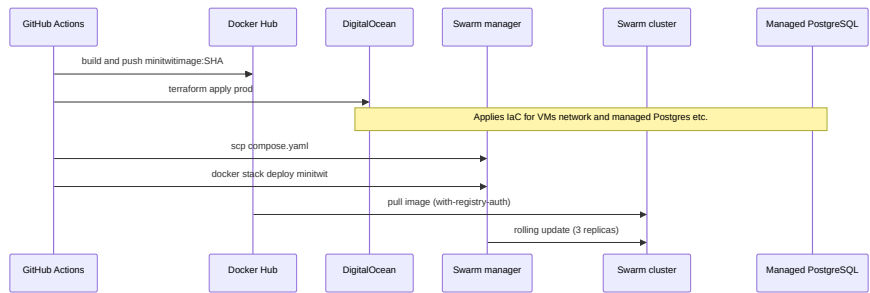


Figure 5: Complete Production Build Workflow

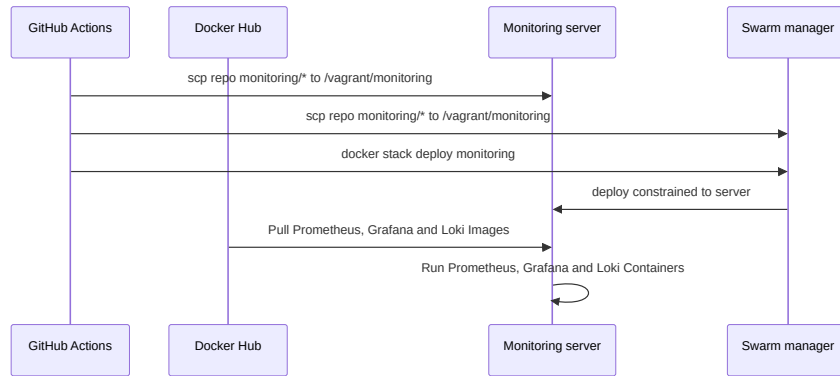


Figure 6: Complete Monitoring Build Workflow

Additional http request metrics are collected through the use of the UseHttpMetrics middleware provided by Prometheus. Custom metric gatherers were also implemented to retrieve metrics from the application's database.

Grafana then retrieves these exposed metrics provided by Prometheus and allows for the construction of various visualizations. We also implemented a Grafana alert based on the up-time metric to inform us when the server was down.

Monitoring Panels

- Current and uptime of container status
- CPU utilization
- Memory utilized by dotnet processes, as well as total physical allocated memory
- Total tweets and tweet rate
- Total users and registration rate
- Http request response latency by their action
- Http GET and POST request rates over time by their response status codes

The following images are examples:

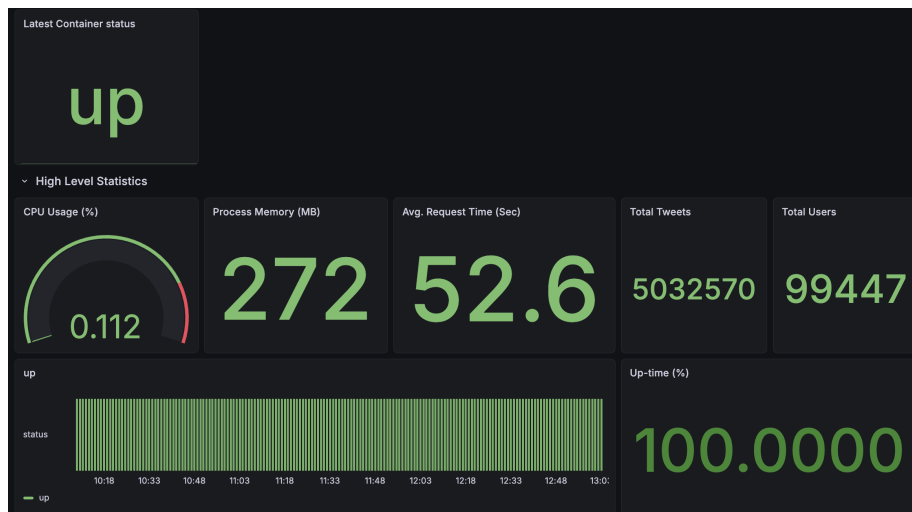


Figure 7: General Statistics

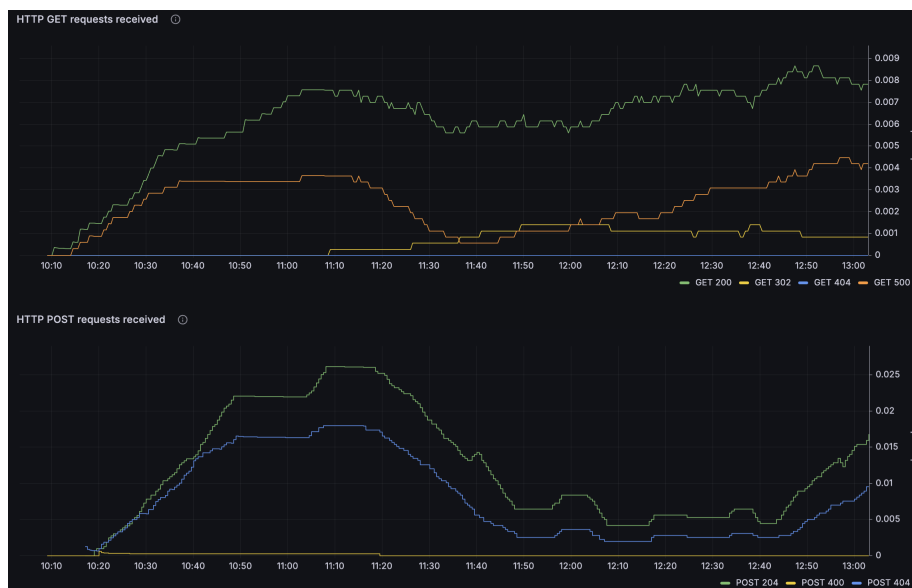


Figure 8: HTTP Requests

2.3 Aggregated logs

We log EF Core's queries and collect them through the built-in Docker functionality `docker logs` for each container. These logs are scraped from all running containers by `promtail`, and then indexed and prepared for presentation in Grafana by `loki`.

The logs are aggregated in Grafana in two dashboards — [PROD](#) displaying data from three deployment replicas and [DEV](#) providing insight into a separate replica used to validate pull requests. All logs are visible side by side in a dedicated logging segment in the Drilldown section, as shown in the image below.

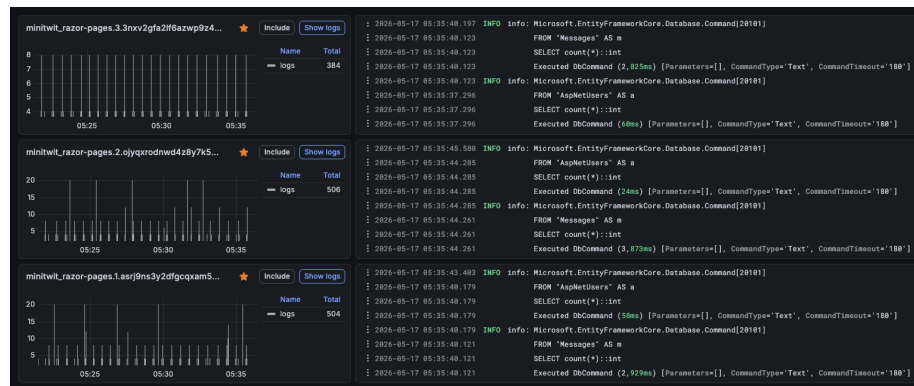


Figure 9: Drilldown Logging Segment

2.4 Security Hardening

For the security hardening of our system a security assessment was made showing an overview of assets, threats, and risks:

Assets - Web Application & API Endpoint - PostgreSQL Database - Monitoring (Grafana, Loki, Promtail & Prometheus)

Threat Sources - SQL Injection - Cross-Site Scripting (XSS) - DDoS Attack - Brute force attack

Risk Scenarios - An attacker uses SQL injection on the web application to gain access to the database. - An attacker injects a script into a post, which is executed on other users' browsers, allowing the attacker to steal session tokens. - An attacker uses a script to spam an API endpoint, resulting in a denial of service.

Risk Analysis

Scenario	Likelihood	Impact	Risk
SQL Injection	High/Common	High	Critical
Cross-Site Scripting (XSS)	High/Common	High	High
DDoS Attack	Medium/Uncommon	Medium	Medium

To solve the various risk scenarios the following measures were taken: - SQL Injection: All inputs are sanitized to avoid script injection. This is handled automatically by Entity Framework Core. - Cross-Site Scripting (XSS): This gains from the input sanitation, but still needs an output encoding to ensure data is rendered as text. Which is handled by Razor Pages rendering all posts as plain text. - DDoS Attack: Access is restricted to only allow a certain amount of requests per/minute, to minimize the effect of DDoS attacks.

Other Security Measures - Setting up inbound firewall rules on DigitalOcean and utilizing `ufw` on the server only allowing specific traffic through on specified ports. Docker does not bypass DigitalOcean's firewalls.

The production application was given firewall rules for: - Standard internet and access ports (TCP 22, TCP 80, TCP 443) - Docker port (TCP 2376) - Docker Swarm infrastructure ports (TCP 2377, UDP 4789, TCP/UDP 7946) - Application ports (TCP 8080) - Grafana Loki log aggregation port (TCP 3100) - Prometheus ports (TCP 9090, TCP 9095, TCP 9096, TCP 9100)

Other security measures were also taken such as:

- Ensuring the application runs on HTTPS with a TLS certificate and setting up `Nginx` for a reverse proxy in front of the application.
- Docker images were also hardened for security by ensuring only user privileges.
- Setting CodeQL up in the repository to scan the code for security vulnerabilities. The static analysis tool automatically discovers source code languages in the repository and dynamically adjusts the scan based on the languages present. CodeQL analyzes the following files in the repository:
 - C# files
 - Python files
 - GitHub Action files
- A Docker image vulnerability scanner, Docker Scout, has been added to the CI workflow to ensure any image vulnerabilities are detected before deployment.

2.5 Availability and Scaling

Availability and scaling in the MiniTwit application are managed by Docker Swarm. A Swarm cluster of the DigitalOcean Droplets is joined into a single Swarm cluster, which continuously monitors and enforces the declared desired state.

High availability is handled by having manager redundancy, three production container replicas, and automatic self-healing. All three nodes in the cluster are given the **manager** role to prevent a single point of failure if one of the manager nodes fails. When a node in the cluster crashes, the Swarm detects a difference between the actual state and the declared desired state, such that the number of actual running replicas is lower than three, which triggers *self-healing* to restore the third replica.

Scaling is handled by Docker Swarm’s built-in Ingress Routing Mesh, which functions as a load balancer. Swarm evenly distributes incoming user requests across all three healthy replicas of the production container to handle high amounts of concurrent requests. This parallelizes the workload across the nodes, such that a container does not consume all the resources of a single node.

To ensure low downtime during the transition from the standalone containers to a Docker Swarm cluster, the stack was deployed with a **blue-green service deployment** strategy from the start, such that the running containers were gradually replaced by the updated ones. This is achieved by configuring the deployment settings within the Docker Compose file to set the update order to **start-first** and a delay parameter that dictates how long Docker Swarm should wait after starting an updated container before terminating an old one, allowing the new container to initialize.

As a result, the services remain available during deployment. By default, Docker Swarm uses the rolling update strategy which terminates the old container before starting a new one. This is referred to as **stop-first**. By terminating the containers first, the default strategy forces the application to experience downtime during the window between container termination and container initialization.

While our strategy should have ensured low downtime during the transition to Docker Swarm, the production application still experienced downtime due to overlooked human errors. These errors came as a result of debugging separate Docker Swarm migration issues. Specifically, the Swarm Ingress routing mesh overwrote the port of the development environment on one of our DigitalOcean Droplets, and the Loki logs failed to display on our Grafana dashboards. During the debugging process, accidental downtime of the application was introduced when pulling the wrong container image due to misconfigured environment secrets, or changing the application to use another port, which prevented the simulator from reaching it.

To avoid these issues in the future, a solution could be to replicate the Docker Swarm infrastructure within an isolated development environment, so any configuration changes during the transition does not affect live production.

3. Reflection Perspective

3.1 Evolution and Refactoring

On the first refactoring from Python Flask to C# RazorPages we ran into unforeseen issues with the methods not working as intended, which slowed us down and required further unexpected bug-fixing before making the release. We had no issues Refactoring to the Onion Architecture. It was time-consuming, but with half of the group being familiar with the framework, the process was relatively smooth.

We discussed defining the infrastructure in Terraform at the beginning of the project, which might have led us to avoiding the amount of “Click-Ops” we had during the project (setting up a managed database, modifying network rules for droplets, etc.), resulting in better reproducibility and version history.

3.2 Operation

We rarely ran into operation issues per se. The system ran without errors most of the time from when the simulator started. For robustness, we set up a QA deployment on Pull Requests, requiring the application to be built, pushed and having its tests pass before merging it into the main branch. This allowed us to test all of our features fully before releasing them, and thereby decreased the amount of bugs and operational work. The experienced downtime was rather due to the deployment of new functionality which didn’t go as planned, such as Docker Swarm or Terraform.

Once we had fully migrated to Swarm, including log shipping from all our Droplets, the Droplet containing the monitoring application ended up being overloaded such that our monitoring application became unreachable. This warranted an upgrade of the Droplet containing the monitoring application. If we had the capacity to provision additional Droplets, we may have considered horizontal scaling instead of vertical. The monitoring Droplet was resized using Terraform and therefore had minimal possible downtime. The full process took ~6 minutes:

Activity

Name	Initiated	Execution Time
 Power on	12 seconds ago	10 seconds
 Resize	1 minute ago	1 minute and 4 seconds
 Power off	6 minutes ago	5 minutes and 4 seconds

Figure 10: DigitalOcean monitoring droplet resize (duration ~6 minutes)

At one point, an unintended addition of a flag reset the volumes for Loki and

Prometheus. After realizing the issue and looking at a few different combinations of flags, we fixed the problem and accepted the loss of earlier metrics & logs. An improvement of the monitoring deployment could be to automatically trigger it on changes to the monitoring folder instead of relying on a manual trigger.

In early April we started receiving warnings from the built-in resource alert system in Digital Ocean that our Database Cluster was above 90% CPU utilization. We started investigating the issue and realized that the amount of requests coming in had ramped up so much that our database could not follow along. We chose to resize the cluster such that it had an extra virtual CPU after a cost-benefit analysis concluding that the developer time it would take to improve the ORM system to send fewer requests would be too time consuming versus the cost of upgrading the database cluster. The database was resized with no downtime.

3.3 Maintenance

The most prominent maintenance issue was arguably the Loki logging instance that repeatedly stopped working for no apparent reason, not affecting operations but hindering development insights. It was fixed multiple times, and seemed to be working for a while, only to crash again at some point. This was a long-lasting debugging task that stretched into weeks as the cause was not immediately clear, with it being a mix of network communication issues, misconfigured ports and an environment name mismatch.

Another hiccup we encountered was related to a fluke in the connection between Prometheus and Grafana, causing an alert to be fired repeatedly. The issue went away after redeployment, and we couldn't definitively arrive at the underlying cause.

There were cases where a functionality developed by one member experienced errors with no clear explanation, only for it to be discovered later that it was caused by a new functionality developed by another group member. One example of such case was when API tests suddenly started failing due to a security measure allowing only a certain amount of requests to reach the server from a single IP, preventing DDoS attacks.

As preventive maintenance, we updated the workflow action versions to be compatible with the upcoming GitHub Actions runtime upgrade to Node.js 24, which issued deprecation warnings.

3.4 DevOps Approach

We described the “DevOps” style of our work extensively in the project's README.md. Overall, we were much more structured and organized in this work compared to previous projects. We made use of the Trello board to track our progress, made sure the sizing of tasks made them self-contained, added a log.md to track our work, extensively tested our code through workflows, and insisted on rigorous code reviews as well as a self-check checklist to follow for

each pull request, providing continuous feedback to group members. We didn't blame anyone for mistakes, but rather offered help and treated each doubt as an opportunity for growth.

4. Use of Generative AI

The use of generative AI tools varied among group members, and so we decided to include separate passages of its use, written by each group member. During groupwork, when we shared the screen and worked on code together, the tool preferred by the person leading the session was used.

Mie mostly consulted the generative AI tool Cursor(<https://cursor.com/>) to discuss issues and warnings throughout the project and provide guidance on issues that we had not encountered before. This was beneficial to the development process as it often unblocked progress, resulting in time savings. Cursor was also helpful in improving the code in terms of maintainability by standardizing its format and structure. It worked quite well together with industry standard formatting / linting tools, making us effectively use a mix of CLI tools and GenAI for improving the code quality in the project. Cursor was also very helpful in summarizing branch work in our log.md and other documentation, ensuring that the chore tasks were actually being done rather than neglected. These logs and documentation written for internal use often reminded us of the work and decision-making performed at different stages of the project.

Daniel used Claude (<https://claude.ai>) and ChatGPT (<https://chatgpt.com/>) to quickly bounce ideas off of it and to help identify the pros and cons of development options with many of them presented. Additionally, both tools were very useful in quickly parsing large error logs.

Mads used ChatGPT (<https://chatgpt.com/>) to debug GitHub Actions, Docker, and DevOps setup issues. He found that it often hindered the work on the deployment workflows, yet helped significantly with commands used in the terminal.

Chris mainly employed GitHub Copilot (<https://github.com/features/copilot>) as interactive documentation to assist in programming. It was also used to propose code changes for very specific improvement cases, usually hinted by SonarQube, such as `b6f71cf3242a25a0c03cd0c0763040417532838f - add wheel hashes to the requirements file`. This made it possible to actually implement security and maintainability solutions which sometimes felt like an overkill.

For Patrick, ChatGPT (<https://chatgpt.com/>) and GitHub Copilot (<https://github.com/features/copilot>) were used to aid the understanding of code errors and thereby helped in solving them. They also showed their utility in queries about writing specific things in different languages, for example: *“How do I write inline code in .md?”* or *“How do I change the rejection status code on the rate limiter?”*

Finally, for Tien, the Google Gemini(<https://gemini.google.com/>) was the primary consultation source used for debugging and resolving technical uncertainty during development.

This included asking the generative AI model to:

- find possible fixes for errors, as well as explaining how and why they appeared.
- provide an overview and understanding of important features and commands of new tools and technologies.
- review developer decisions to ensure that any changes to the application is correct.
- check grammar and phrasings when writing documentation or report.